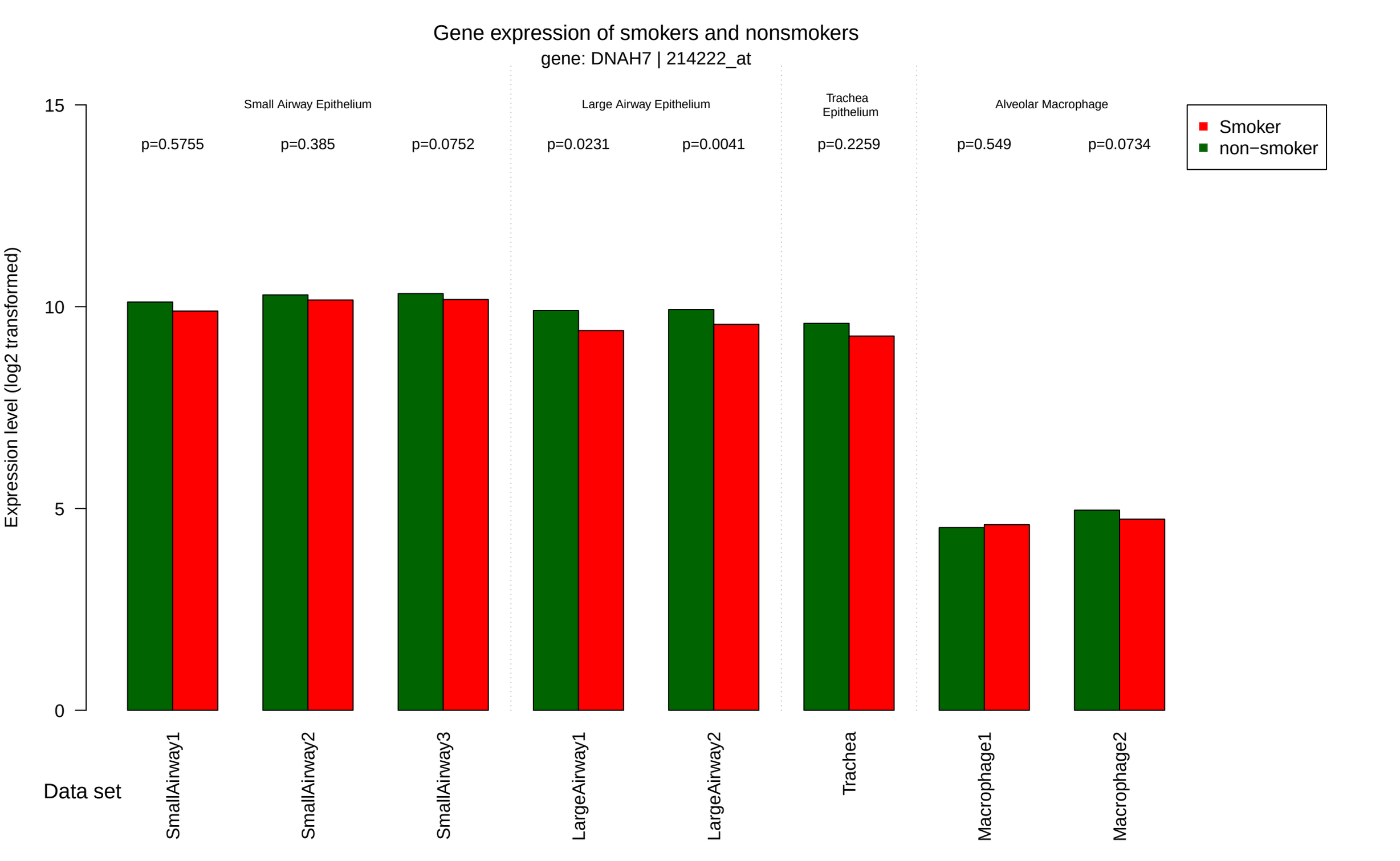
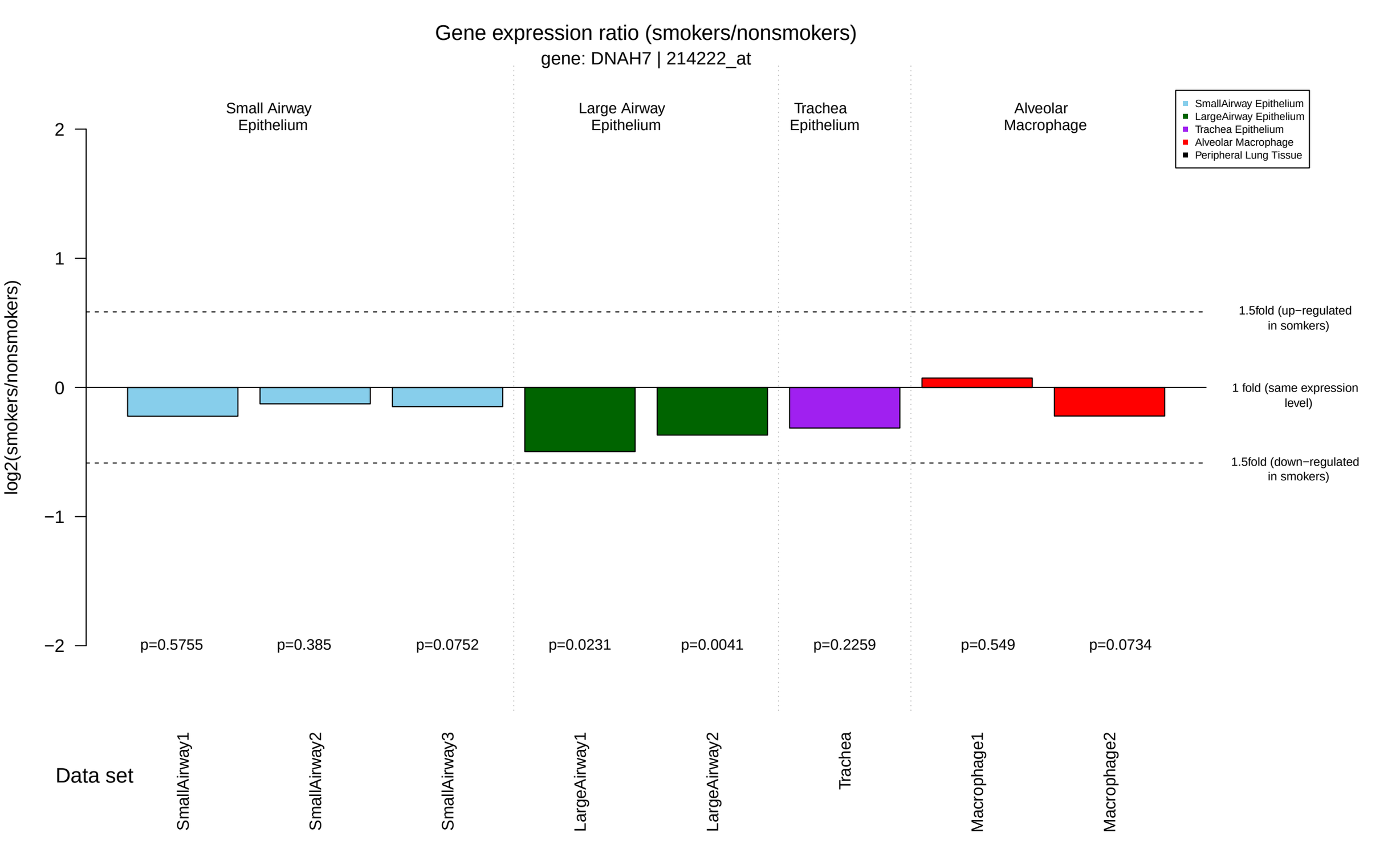
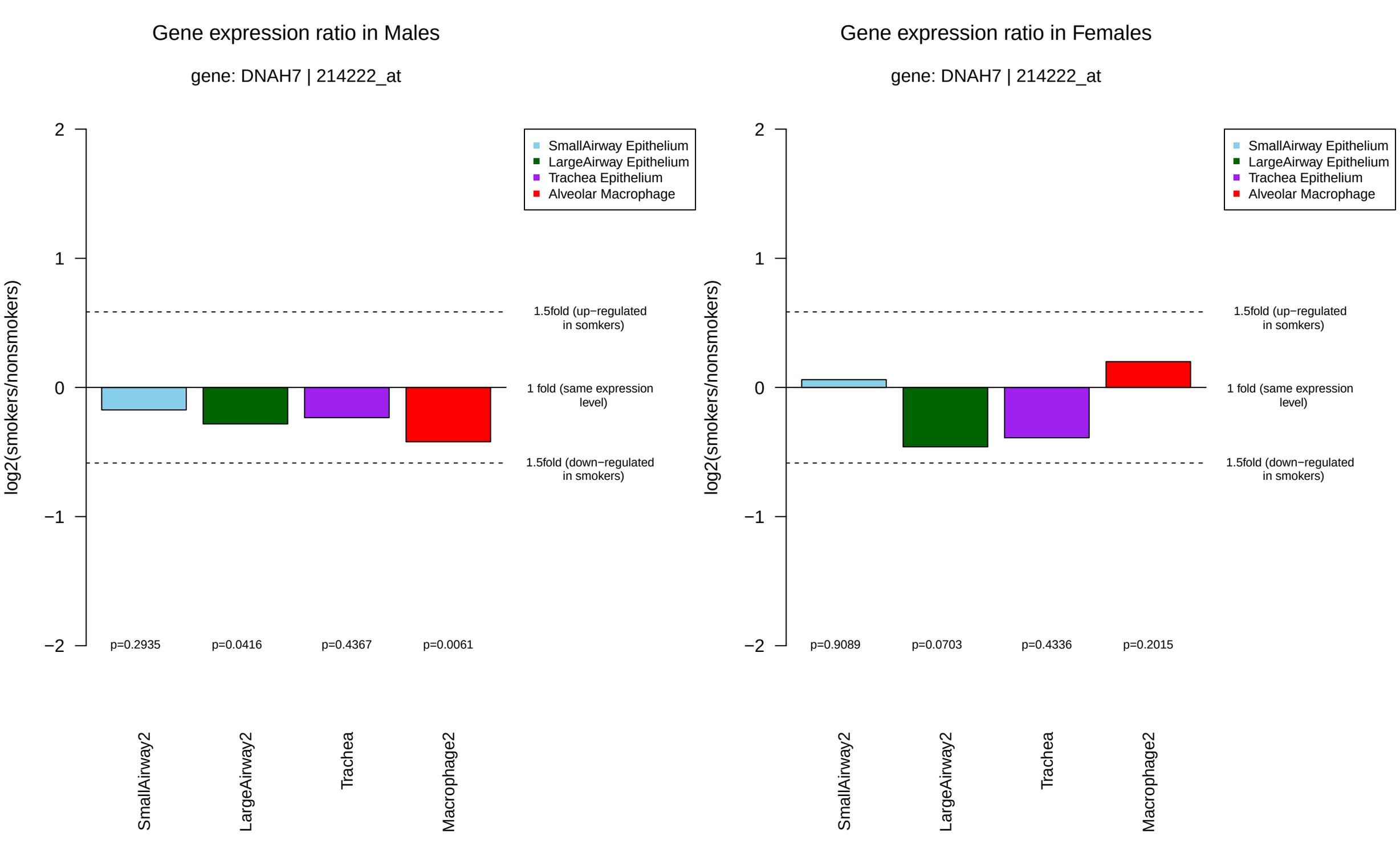
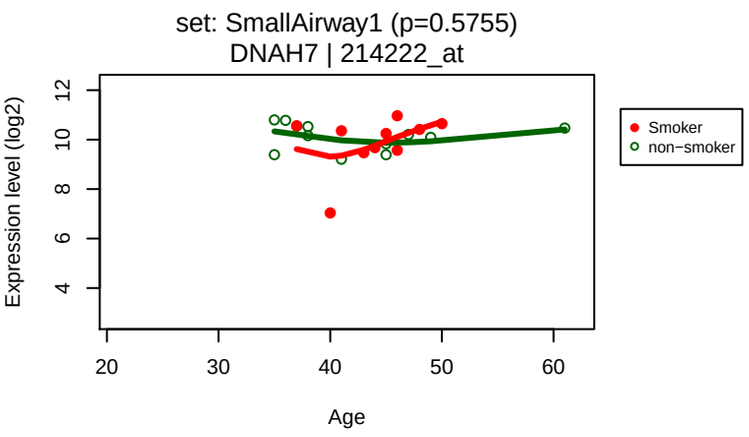
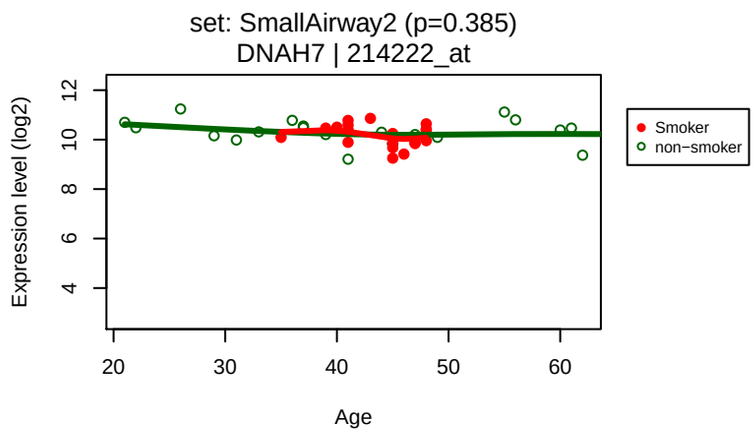
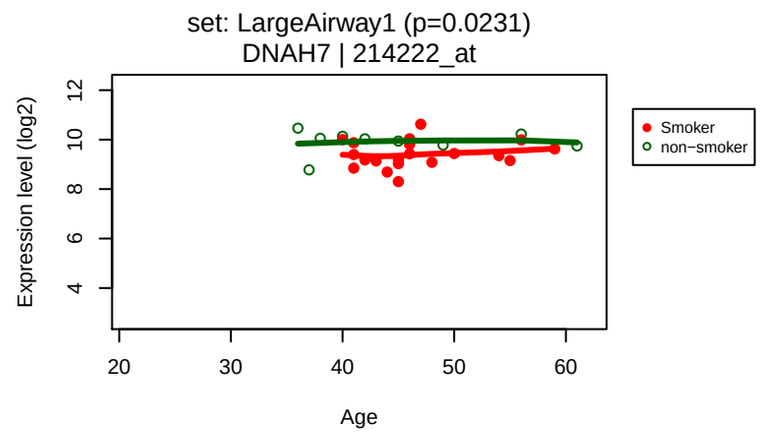

Gene expression of smokers and nonsmokers
gene: DNAH7 | 214222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5755
p=0.385
p=0.0752
p=0.0231
p=0.0041
p=0.2259
p=0.549
p=0.0734
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

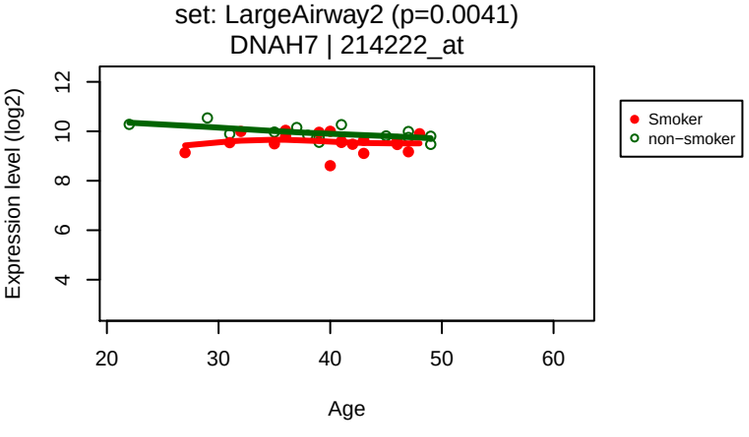

Gene expression ratio (smokers/nonsmokers)
gene: DNAH7 | 214222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5755
p=0.385
p=0.0752
p=0.0231
p=0.0041
p=0.2259
p=0.549
p=0.0734
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

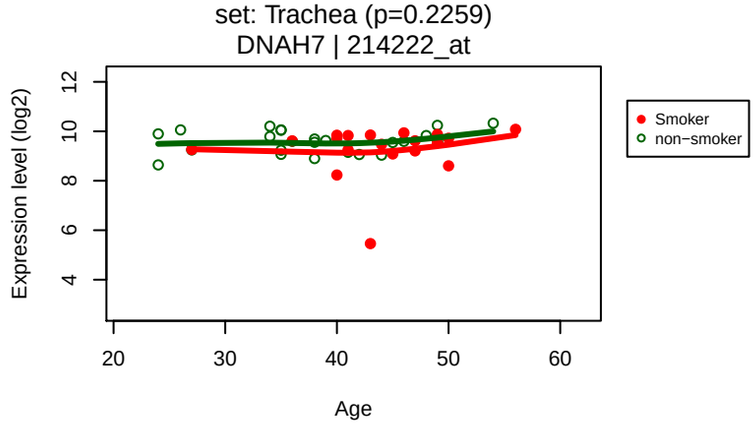

Gene expression ratio in Males
gene: DNAH7 | 214222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2935
p=0.0416
p=0.4367
p=0.0061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DNAH7 | 214222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9089
p=0.0703
p=0.4336
p=0.2015
SmallAirway2
LargeAirway2
Trachea
Macrophage2

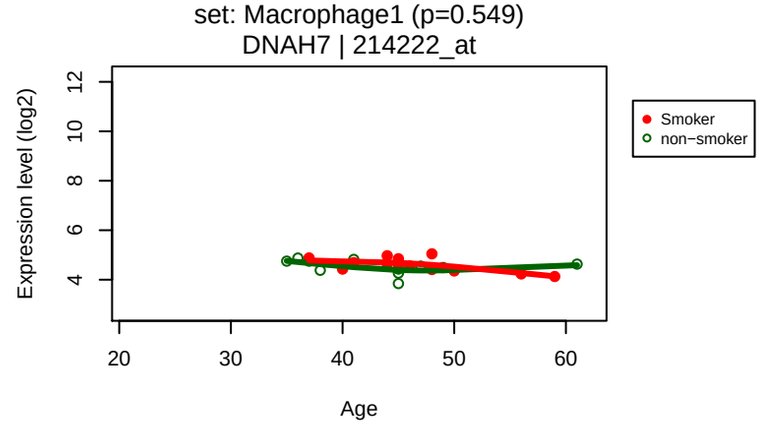

set: SmallAirway1 (p=0.5755)
DNAH7 | 214222_at
Expression level (log2)
Age
Smoker
non−smoker

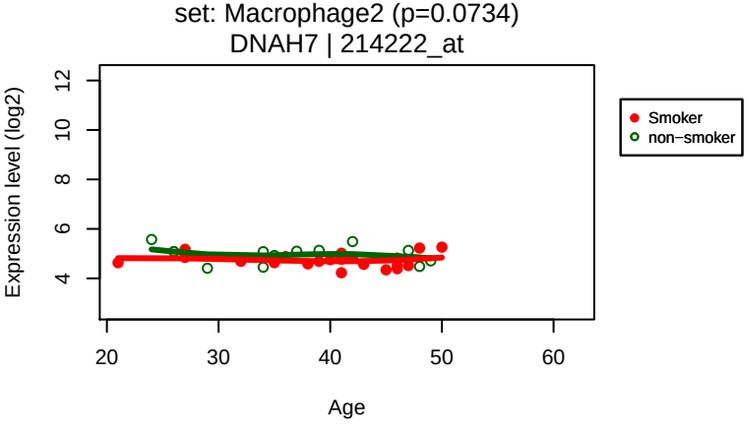

set: SmallAirway2 (p=0.385)
DNAH7 | 214222_at
Expression level (log2)
Age
Smoker
non−smoker

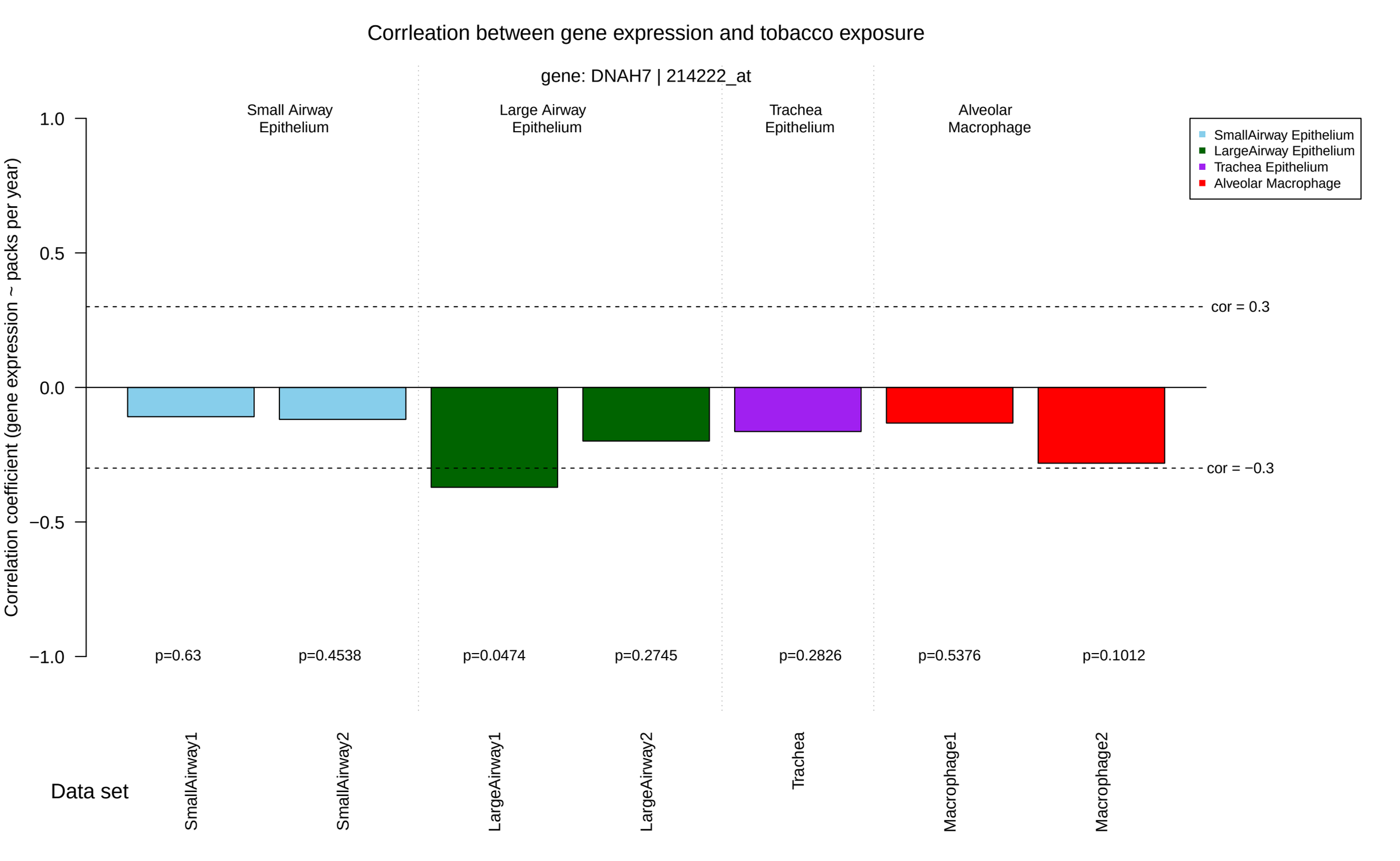

set: LargeAirway1 (p=0.0231)
DNAH7 | 214222_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0041)
DNAH7 | 214222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2259)
DNAH7 | 214222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.549)
DNAH7 | 214222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0734)
DNAH7 | 214222_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DNAH7 | 214222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.63
p=0.4538
p=0.0474
p=0.2745
p=0.2826
p=0.5376
p=0.1012
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2